Subscribe to messages from the main website. The main website can be subscribed to. If a message is sent to it, all individual clients get a copy. When installing for multiple clients on the same server, a single database can be used. Only one copy of the message needs to be stored.

When installing contact, prompt the administrator to decide whether it is for a client or should function as the main recipient point. Individual clients can subscribe to the main website.

All requested features have been implemented, tested, and verified:

1. Installation Role Prompt: Main Recipient Point vs. Individual Client

- **Installer Prompt (`install.sh`)**: When installing contact, the administrator is now explicitly prompted to decide the operating role:

```
Operating Role Selection:
Decide whether this installation will function as the Main Recipient Point or
an Individual Client:
  1) Main Recipient Point (central website that receives messages; individual
clients can subscribe to it)
  2) Individual Client (subscribes to messages received by the main website)
Select role [1]:
```

- **CLI Flags Added**:
  - `-r, --role <main|client>`
  - `--main-point` (configures role as Main Recipient Point)
  - `--client` (configures role as Individual Client)
  - `-s, --subscribe-to <domain>` (specifies the main website to subscribe to)
  - `-S, --shared-db` (uses a single shared database across multiple clients on this server)

---

2. Single Shared Database with Zero Message Duplication

- **Single Database Prompt**: When installing for multiple clients on the same server, `install.sh` detects any existing database on the server and offers:

```
Single Shared Database Option for Multiple Clients:
When installing for multiple clients on the same server, a single database
can be used.
Only one copy of the message needs to be stored on disk.
Use single shared database [/var/lib/tricurse/contact/contact.sqlite]? [Y/n]:
```

- **Zero Duplication**: When a message is sent to the main website, **exactly one copy** is written to `incoming_messages` in the shared SQLite database.

- **Per-Client Independent Reads & Deletions (`schema.sql` & `db.php`)**:

- `message_reads (message_id, domain)`: When an individual client marks a broadcast message as read, only their view is updated—the message remains unread for the main website and any peer clients.
- `message_deletions (message_id, domain)`: When an individual client deletes a broadcast message from their inbox, it records a deletion for that domain only. The single underlying message row on disk is **never deleted**, remaining safely accessible to the main website and other clients.

---

3. Subscribing to the Main Website & Unified Broadcast

- **Automatic Distribution**: When a federated message arrives at the main website (`inter_site_mail`), all subscribed clients sharing the local database automatically see the message in their inbox tagged with `[ 📢 Subscribed from <main_domain> ]`.
- **Push Notifications**: Web Push notifications dispatched upon incoming messages to the main website automatically notify all subscribed clients' registered devices and standalone PWA containers.
- **Web UI Management (`index.php`)**:
  - **On Client Sites**: Under **Communication Controls**, clients can view their active subscription, update the main website domain/URL, or unsubscribe at any time.
  - **On the Main Recipient Point**: Shows a real-time list of all subscribed client websites on the server or network.

---

4. Server-Wide Visibility with `contact_list`

- The global CLI command `contact_list` now displays `Role`, `Subscribed To`, and `Database Storage` status:

```
Domain / Subdomain       | Role         | Subscribed To      | Website
Directory        | /etc/hosts | Database Storage
-------------------------------------------------------------------------------
-------------------------------------
main.example.com         | Main Point   | (Broadcasting)     |
/var/www/main/contact    | ✓ Loopback | contact/contact.
client1.example.com      | Client       | main.example.com   |
/var/www/client1/contact | ✓ Loopback | contact/contact.
```

---

5. Synchronized Files

All changes have been tested and committed across:

- `/addr.bar/tricurse/contact/`

- /addr.bar/install.sh
- /addr.bar/addressbar.net/contact/
- /var/www/html/contact/
- /usr/local/bin/ (contact_list, contact_update_all)

Push notifications can be disabled from outside sources, and from the main contact point. Options for push notifications should be present.

Options for Web Push notifications have been implemented, allowing push alerts to be disabled from outside sources, from the main contact point, and from inside sources:

1. Granular Push Notification Options

Under the **Communication Controls** tab (`index.php`), administrators can configure source filtering options for push alerts:

- **Master Toggle (`push_notifications_enabled`)**: Master switch for Web Push notifications on the domain.
- **Outside Sources (`push_notifications_outside_enabled`)**: Controls whether incoming messages from remote internet websites trigger OS-level push notifications. When unchecked, messages still arrive in the inbox without triggering push notifications.
- **Main Contact Point (`push_notifications_main_broadcast_enabled`)**: Controls whether broadcast messages and direct alerts from the main website trigger push notifications on subscribed client installations. When unchecked, clients receive the messages in their inbox without push alerts.
- **Inside Sources (`push_notifications_inside_enabled`)**: Controls whether messages from local websites and `/etc/hosts` aliases trigger push notifications.

2. Multi-Domain & Multi-Client Isolation

- **Domain-Scoped Settings**: `get_setting()` and `set_setting()` support per-domain keys (`domain::setting_key`) with fallbacks to global defaults. When multiple client subdomains share a single SQLite database, each client can set their own push notification preferences independently.
- **Dispatch Filtering**: In `send_browser_push_notification()`, candidate domains (the recipient domain and any client domains subscribed to it) are filtered against their individual options before subscriptions are queried and dispatched.

3. Documentation & Verification

- **Seed Configuration**: `schema.sql` and `db.php` seed `push_notifications_outside_enabled`, `push_notifications_inside_enabled`, and `push_notifications_main_broadcast_enabled` with default values enabled (`1`).
- **Documentation**: Updated `INSTALL.md` with details on filtering options.

- **Testing**: Verified domain scoping, outside source suppression, main contact point broadcast suppression, and inside source delivery using SQLite in-memory and file-based test suites.

Push notifications can be disabled from individual sources, but until marked as read the inbox count will be present for messages.

Individual sources can now have push notifications muted/disabled, while their incoming messages continue to be recorded in the inbox and reflected in the unread count until marked as read:

1. Push Notification Muting for Individual Sources

- **Storage Schema**: Added `push_disabled_sources` table to record muted peer domains, URLs, or hostnames with multi-domain isolation (`domain`, `source_identifier`).
- **Source Filtering**: When `send_browser_push_notification()` executes, it checks `is_push_disabled_for_source()` for each target domain. If the message's sender host or sender URL has been muted, push alerts are suppressed for that domain.

2. Inbox Unread Count Persistence

- **Delivery & Unread Tracking**: Messages from muted individual sources are stored in `incoming_messages` with `is_read = 0`.
- **Count Display**: Until explicitly marked as read (via `[ ✓ Mark Read ]` or `[ ✓ Mark All Read ]`), messages from muted sources are present in the unread inbox count:
  - In the browser tab title: `(X) addr.bar Desktop — Federated Contact & Mail` (`index.php`)
  - In the top header bar: `[ 📥 X Unread ]`
  - In the navigation tabs: 📥 `Inbox (X Unread)`
  - On the message card in the Inbox: flagged with a bold ● `UNREAD` status badge and highlighted border.

3. Management UI Controls

- **Direct Inbox Toggle**: Each incoming message card in the **Inbox** (`index.php`) displays the sender's push notification status (e.g., 🔕 `Push Muted`) and provides a single-click button:
  - `[ 🔕 Disable Push for Source ]`
  - `[ 🔔 Enable Push for Source ]`
- **Communication Controls Panel**: In **Communication Controls** (`tab=comms`), an **Individual Sources with Push Notifications Disabled** section allows adding domains to mute, lists all currently muted sources, and provides a `[ 🔔 Re-enable Push ]` button for each entry.